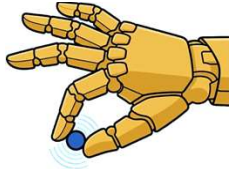




Programmable Magnetic Micro-Cilia for Distributed Haptic Actuation

Swastik Kushagr, Dinesh K. Patel, Rahul Panat
IROS 2026 Workshop: Enhancing Dexterous Manipulation Through Haptic Feedback

IROS 2026
PITTSBURGH



Challenge: Distributed, Programmable Haptic Actuation

Haptic Need

Candidate Micro-Actuator

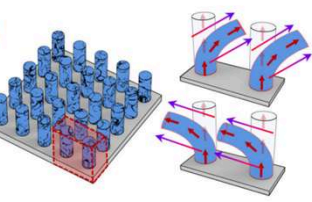
Manufacturing Bottleneck



Shan et al., Adv. Sci. (2025)

Dense actuator arrays enable localized tactile feedback

- High-density actuator arrays for fine spatial resolution
- Localized, independently mechanical stimuli
- Programmable displacement and force

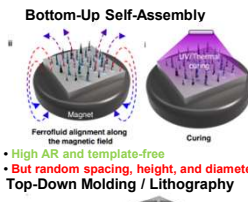


Actuation of artificial magnetic cilia

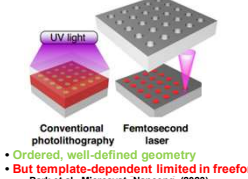
Park et al., Microsyst. Nanoeng. (2023)

Magnetic cilia: remote, compliant microscale actuation

- Remote, contactless magnetic actuation
- High-aspect-ratio cilia
- Compliant bending response



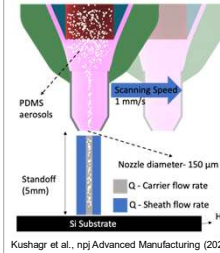
• High AR and template-free
• But random spacing, height, and diameter
Top-Down Molding / Lithography



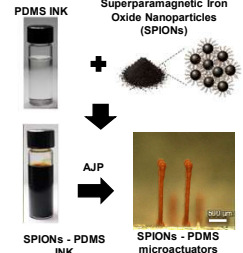
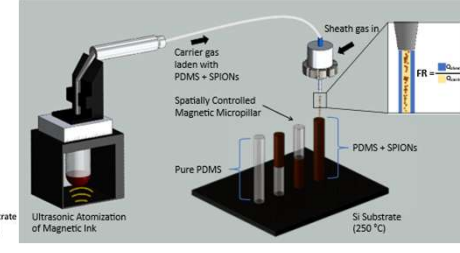
• Ordered, well-defined geometry
• But template-dependent limited in freeform
Park et al., Microsyst. Nanoeng. (2023)

Need: Maskless, deterministic fabrication of high-AR, spatially programmable magnetic micro-actuators

Approach: Multimaterial 3D Aerosol Jet Printing



- Thermally assisted AJP enables support-free vertical PDMS growth
- Two printable materials: pure PDMS + PDMS-SPION composite
- Material switching enables deterministic magnetic/non-magnetic
- Toolpath/layer control enables high-AR, freeform microscale structures

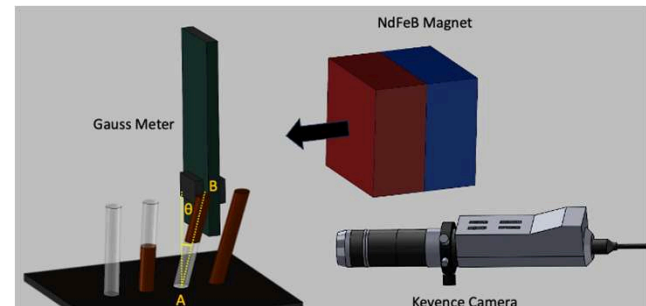


- PDMS (10:1) diluted 10× with mesitylene for aerosolization
- Tunable SPION loading: 7.6, 24.7, and 39.6 wt%
- Stable magnetic-ink dispersion for ≥ 3 weeks

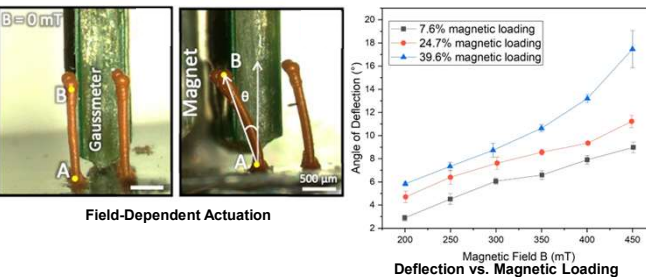
One Platform, Three Knobs: Composition, Placement and Geometry

Programmable Actuation Through Composition, Placement, and Geometry

Magnetic Loading Controls Actuation

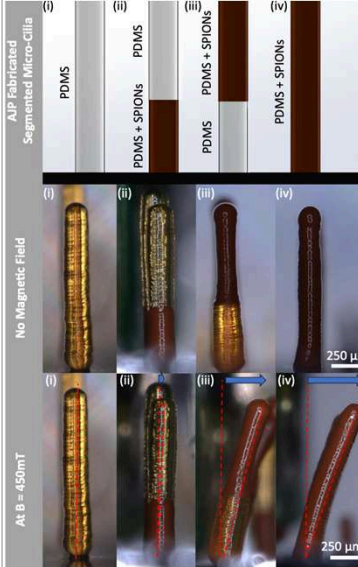


Experimental Setup



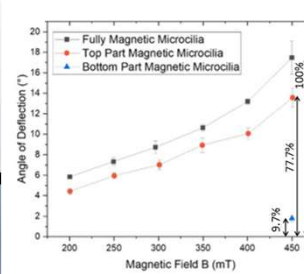
- Deflection increases with magnetic field
- Higher SPION loading increases actuation
- 39.6 wt% gives ≈2× the deflection of 7.6 wt% at 450 mT
- Composition provides a geometry-invariant tuning parameter

Spatial Material Placement Programs Actuation

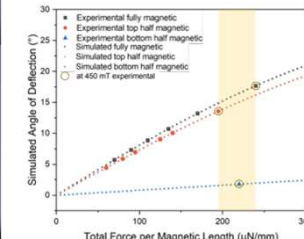


Spatial Material Placement Enables Programmable Actuation

- Identical overall geometry; magnetic material repositioned axially
- At 450 mT: Full 17.5° | Top 13.6° | Bottom 1.7°
- Nearly 10× actuation contrast through material placement

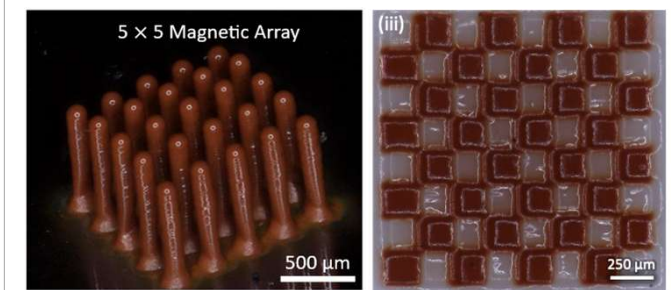


Field-Dependent Actuation by Material Placement



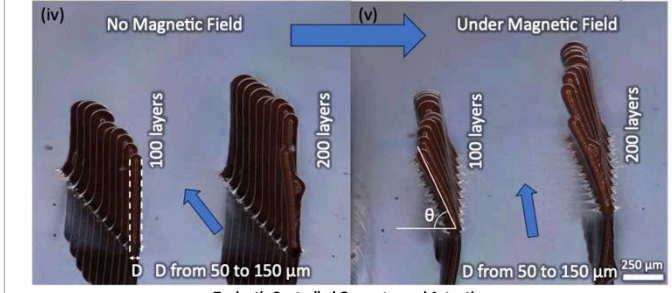
Experiment-Simulation Validation

Geometric Design Freedom & Scalability



Scalable 5 × 5 Magnetic Micro-Cilia Array

Multimaterial Spatial Patterning



Toolpath-Controlled Geometry and Actuation

- Support-free inclined magnetic micro-cilia
- Diameter control from 50-150 μm
- Multimaterial spatial patterning
- Ordered 5×5 magnetic micro-cilia array